

Video Deflector Modification

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REVISION HISTORY

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A	2000-06-06	Draft	N.G. Johnson
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1.Introduction

The existing gun produces a pulsed electron beam with an energy of 220 keV. An electrostatic chopper, called the video deflectors, rapidly deflects the beam through a collimator decreasing the rise time of the leading and trailing edge of the pulse, and defining the pulse length. There has been an increased interest in operating the CLS in single bunch mode. This will either require a new gun, or a much faster video deflector system.

2.Scope of Work

Driver electronics based on avalanche transistors will be developed. The configuration of the video deflector plates must also be redesigned to include the effects of reflections from the much faster slew rates.

2.1Video Deflector Amplifier

The video deflector system consists of two parallel plates, and a dipole magnet, both capable of deflecting the beam in the horizontal plane. With no potential difference between the plates the magnet will deflect the beam away from the center of a collimator. The proper voltage difference between the plates will cancel the magnetic deflection, and the beam will pass through the center of the collimator. Initially the plates will both be charged to a high positive voltage. As the first deflector is triggered the plate will be pulled to ground potential, developing the necessary electric field to deflect the beam through the center of the collimator. After the second deflector is triggered both plates are at ground potential, and the dipole field deflects the beam out of the collimator.

A 500 MHz chopper is under development. It will be located near the gun, and is being designed to chop the beam to less than 130 degrees, or 0.7 nS. This leaves the gap between pulses, or 1.3 nS, for the deflector system to bend the beam in or out of the path through the collimator. A portion of this time is required for errors such as the incremental delay steps of the timing system and jitter of the deflector amplifier electronics. The goal of the amplifier will be a pulse which switches the beam through an on-off or off-on transition in less than 1 nS.

By using additional voltage, the amplifier can meet this goal even if its rise time is more than 1 nS. For example, the existing deflector plates require approximately 200 V to deflect the beam into the collimator. If the amplifier drove the plates with 600 V linear ramp, the rise time of the ramp could be 3 nS. As the ratio of the applied voltage to the required voltage increases, ringing has an increasingly detrimental effect, since ringing is generally proportional to the size of the transition.

2.2Video Deflector Plates

The deflector plates must be considered as a transmission line. Ideally the deflector plates and vacuum feed through assembly would match the driver impedance. The configurations being considered are presented in Appendix B.

3. Final Specifications, Plans and Details

The video deflector plates will be designed so a voltage of 200 – 300 V deflects the beam from intercepting the collimator to the middle of the collimator. The driver circuit is designed to provide a 750 V leading edge, which results in faster switching since only a fraction of the risetime is required to deflect the beam.

A dedicated oscilloscope has been added for monitoring circuit operation.

The components of the system include:

- Avalanche transistor driver design.
- Video deflector plate design.

- Oscilloscope located near gun for remote monitoring.

3.1Avalanche transistor deflector driver

The schematic, part list, and component placement drawings are included in Appendix B.

3.2Video deflector plate assembly prototype

Mechanical drawings for the prototype deflector plates are included in Appendix C.

4.Final Design Verification

4.1 Driver circuit

The schematic and PCB layout were generated by J.M. Vogt and checked by N.G. Johnson.

4.2 Video deflector plates

The prototype mechanical drawings were generated by J.A. Swirsky and checked by D.R. Yosifov.

The deflector configuration will be finalized based on the prototype test results.

5. Definition of Material

The part lists are included in the Appendices.

6. QA/QC Standards and Process Requirements

The solder joints of all circuits will be visually inspected under a magnifier.

QA/QC standards for the deflector plate assembly will be defined when the final version of the plates is designed.

7. Spares Identification

Two complete driver assemblies will be constructed as spares. One assembly will be available for testing/development without affecting spares.

8. Updated Schedule and Resource Requirements

The Receivers for Video Deflector are needed by December 1. All other devices are needed by the middle of December to allow installation before the planned operation of the LINAC in January 2001.

All cost estimates are additional cost to money already spent.

Task	Start Date	End Date	Engineering	Tech
Design Note		2-Jun	5	
Deflector Plate/Feedthrough Design	2-Jun	31-Dec	10	5
Driver Design	2-Jun	15-Nov	30	30
Vacuum Design	15-Oct	15-Nov	5	5
Order PCB	15-Nov	7-Dec	1	1
Machine Final Components	15-Nov	31-Jan	2	5
Assemble	1-Feb	15-Feb	2	2
Install	1-Feb	28-Feb	2	2
Test/Approve	1-Mar	15-Mar	6	5
Manual/Documentation	1-Dec	30-Mar	5	1

Forecast additional cost:

Item	Price
PCBs	\$700
Deflector assembly	\$2000
Oscilloscope	\$2000
Total	\$4700

Appendix A – Video Deflector Driver

Drawing number ACCL/EE/VD/0046100

- Schematic
- Partlist
- Component Placement
- Bottom Component Placement
- Fiber Optic Receiver Chassis

Appendix B – Video Deflector Plates

Drawing number TST/ME/JIG/0047000 - 0047004

- Test Setups and Parts lists

Drawing number TST/ME/JIG/0047011 - 0047023

- CLS Machined Parts